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Superseding
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MILITARY SPECIFICATION

CENTRAL OFFICE TELEPHONE, MANUAL AN/TTC-23( )

1. SCOPE

1.1 Scope.- This specification covers Central Office, Telephone, Manual AN/TTC-23( ) which is a transportable system capable of terminating 100 local and 20 trunk telephone lines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equip.)
MIL-W-6858	Welding, Resistance
MIL-T-7928	Terminals, Lug and Splice, Crimp-Style, Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-55505	Shelter, Electrical Equipment (containing equipment) Packaging and Packing of

STANDARDS

MIL-STD-1235	Single & Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC - 5805

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MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment,Classification of Visual and Mechanical Defects

DRAWINGS

ELECTRONIC COMMAND

SC-DL-582402

(Copies of specifications,standards,drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Central Office, Telephone, Manual AN/TTC-23() includes modified Shelter S-280()/G and the following major installed items:

2 each Switchboard,Telephone,Manual SB-1398()/GTA-14(V)
1 each Panel,Power Distribution SB-1399()/GTA-14(V)
1 each Telephone Circuit,Line Relay TA-452()/GTA-14(V)
2 each Main Distribution Frame TA-454()/GTA-14(V)
1 each Telephone Circuit,Trunk Relay TA-276()/TTC
6 each Headset-Microphone H-210()/G
2 each Telephone Set TA-312()/PT
1 each Intercommunication Station LS-147()/F1

Central Office,Telephone,Manual AN/TTC-23() requires 115 VAC,60 cycle, single phase primary power. Field power can be supplied by Generator Set, Gasoline Engine, Trailer Mounted PU-618()/M, which is not part of the AN/TTC-23().

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3.2 Preproduction inspection.- The contractor shall subject the first article of the AN/TTC-23() equipments on order to the preproduction tests specified and shall submit the tests results, in the form of a report, for approval if required by the invitation for bids and contract. (See 4.3 and 4.6)

3.3 Construction.- The Central Office, Telephone, Manual AN/TTC-23() shall be constructed in accordance with drawings listed on SC-DL-582402. (See 4.19)

3.4 Electrical installation.-

3.4.1 Power wiring.- The power wiring shall be in accordance with the drawings and meet the following requirements:

3.4.1.1 Breakdown voltage, insulation.- Electrical insulation shall withstand a 10-second application of 500 VDC. The leakage resistance shall not be less than 100 megohms. (See 4.6.1.1)

3.4.1.2 Operation of distribution panel circuits.- All components on all circuits of the distribution panels shall meet continuity and operational tests as follows:

Power Distribution Panel, Circuit Breakers No.1 through No. 6. (See 4.6.1.2 through 4.6.1.7)

Emergency Light Tests. (See 4.6.1.8)

3.4.2 Signal wiring.- The signal wiring shall be in accordance with the drawings and withstand a 10-second application of 500 VDC. The leakage resistance shall not be less than 100 megohms. (See 4.6.2)

3.4.3 Cord assemblies.- All telephone cord assemblies and air conditioner cable assembly shall be in accordance with Specification MIL-C-3885.

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3.5 System operation.- With all operating components of the Central Office, Telephone, Manual AN/TTC-23() installed as indicated by the drawings, the equipment shall function in accordance with the following requirements:

3.5.1 Power panel SB-1399()/GTA-14(V).- With all equipments associated with the power panel connected for operation, the power panel shall be capable of furnishing necessary voltages and ringing power for the switchboards. (See 4.6.3.1)

3.5.2 Subsidiary switchboard functions.- The following subsidiary switchboard functions shall be capable of being performed without fault:

Supervisory Recall Lamp (See 4.6.3.2)
Dial (See 4.6.3.3)
Hand Ringing Generator (See 4.6.3.5)
Mon-Master Ringing (See 4.6.3.6)
Position Grouping (See 4.6.3.7)
Monitor (See 4.6.3.8)
Conference Call (See 4.6.3.9)

3.5.3 CB-LB Line circuit operation.- With all operating components connected as required by the drawings, communication through the switchboard shall be possible. (See 4.6.3.4 and 4.6.3.10)

3.5.4 Information turret operation (Info phones).- With all operating components connected as shown in the drawings, the information turret shall be capable of receiving ahd holding two subscriber phone calls at the same time. (See 4.6.3.12)

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3.5.5 Trunk circuits operation.- With selected switches of Trunk Telephone Circuit TA-276()/TTC in their ON position, sequential operation of the idle trunk circuits shall be possible (See 4.6.3.11)

3.5.6 Intercommunication Station Operation (Supervisor's desk).- With an Intercommunication Station LS-147()/F1 connected to the INTERCOM binding posts in Signal Entrance Box, communications shall be possible with intercom at Supervisor's desk. (See 4.6.3.13)

3.5.7 Supervisor's Phone Operation.- With a telephone connected to the Phone (supervisor) binding posts in the Signal Entrance Box, communications shall be established with the Phone (Supervisor) at the desk. (See 4.6.3.14)

3.5.8 Spare Lines on Jack Panel.- With phones arranged for LB operation, communications shall be possible between spare jacks at Supervisor's position and the connectors in the Signal Entrance Box. (See 4.6.3.15)

3.6 Vehicular transportation.- Central Office, Telephone, Manual AN/TTC-23() shall be capable of being driven over cross-country terrain by military vehicle without sustaining any permanent damage. (See 4.7)

3.7 Railroad transportation.- Central Office, Telephone, Manual AN/TTC-23() shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail. When so loaded, the equipment shall be subjected to and withstand the test of paragraph 4.8 without sustaining any buckling, delamination, or similar damage. Deformation in the ceiling and floor immediately above and below the switchboards, line and trunk relay frames, and main distribution frames shall not exceed 5/32-inch after completion of rail tests.

3.8 Drops, flat and rotational.- Central Office, Telephone, Manual AN/TTC-23() shall withstand drops of 18 inches onto concrete. After each of the drop tests of 4.9.1 and 4.9.2, there shall be no buckling, delamination, or similar damage to the shelter. Deformation in the ceiling and floor immediately above and below the switchboards, line and trunk relay frames, and main distribution frames shall not exceed 3/32-inch after completion of all drops. Deformation of battery acid tray shall not exceed 1/8-inch after completion of all drops.

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3.9 Watertightness.- Central Office, Telephone, Manual AN/TTC-23( ) shall withstand the test of 4.10. After the tests, there shall be no leakage of water into the shelter.

3.10 Fording.- Central Office, Telephone, Manual AN/TTC-23( ) shall be capable of being immersed in water to a depth of 21-inches, measured from the bottom of the shelter skids, for a period of one (1) hour. Upon completion of the test there shall be no leakage of water into the shelter. (See 4.11)

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of auxilliary materials, such as adhesive, and when installed each insert shall be capable of withstanding the torque loads referenced in paragraph 4.13. There shall be no damage to the insert or insert threads. The insert shall not have rotated from its original position, and there shall be no distortion of the shelter mounting members, the ceiling, wall, or floor areas immediately surrounding the insert being tested.

NOTE: Verification that the inserts have met the requirements of paragraph 4.13 shall be accomplished before the inserts are used to fasten bracketry, racks, etc. Verification shall be witnessed by a Government representative.

3.12 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.13 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings.

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3.14 Marking.-

3.14.1 General.- Marking shall conform to Specification MIL-M-13231 and the equipment drawings.

3.14.2 Nameplates.- Each Central Office, Telephone, Manual AN/TTC-23( ) shall be provided with a nameplate. Each nameplate shall be imprinted with a sequential serial number.

3.14.3 Aircraft loading data plate.- Each Central Office, Telephone, Manual AN/TTC-23( ) shall be provided with a data plate in accordance with the drawings.

3.14.4 Interior and exterior marking.- All interior and exterior marking of the equipment shall be as depicted on the drawings.

3.15 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.16 Seal, vapor.- All joints and raw edges of the framing for the blower exhaust, power and signal entrance boxes, and air conditioner cutouts shall be sealed to provide a vapor barrier against the entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall have the properties of 3.16.1, 3.16.2, and 3.16.3. (See 4.14)

3.16.1 Sealer shear at temperature.- The minimum shear strength of the fully cored sealer and its bond to the aluminum shall be as follows:

150 psi at ambient ($80^{\circ}F \pm 15^{\circ}$)
150 psi at $-65^{\circ}F$
75 psi at $200^{\circ}F$

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3.16.2 Sealer shear after humidity exposure.- The sealer shall withstand exposure to 100 percent relative humidity at 160° F for 2 weeks and have a shear strength of 75 psi minimum.

3.16.3 Sealer shear after salt spray exposure.- After 2 weeks exposure to 20 percent NaCl solution at 95° F the sealer shall have a shear strength of 150 psi minimum.

3.17 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the Central Office, Telephone, Manual AN/TTC-23() securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified transport tests.

3.18 Soldering.-

3.18.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.18.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.18.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The soldering shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the

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equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.19 Tropicalization of material.- Canvas,webbing; wood etc. shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.19.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.19.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.19.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.19.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use for materials such as canvas, webbing, cloth, etc., cited on drawings listed on SC-DL-582402. (See 6.2(d)).

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3.20 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.21 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.21.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided.

(a) To prevent undue stress on cable forms, wire, and connections.

(b) To facilitate field repair of broken or cut wires.

(c) To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.21.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.21.3 Splicing.- Wires in continuous run between two terminals shall not be spliced during the wiring operation.

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3.21.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements of Specification MIL-T-7928. (See 4.4 and 4.12.)

3.21.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.22 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.15.) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.23 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.3 Construction
3.12 Cleaning
3.13 Finish
3.14 Marking
3.15 Riveting
3.16 Seal,vapor
3.17 Securing of parts
3.18 Soldering
3.19 Tropicalization
3.20 Welding
3.21 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery.) (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.17.)

4.3 Preproduction inspection.- This inspection will be performed by the contractor on the First Articles unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (see Tables II,III, and IV, respectively). The preproduction inspection will normally be performed in this order:

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a. Mounting inserts
b. Power wiring
c. Signal wiring
d. System operation
e. Vehicular transportation
f. Rail transportation
g. Drop tests
h. Watertightness
i. Fording

The last tests performed shall be a repeat of the Systems operation tests. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table I - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req't Para	Insp Para
Vehicular transportation	3.6	4.7
Rail transportation	3.7	4.8
Sealer, Shear	3.16	4.14

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Cord Assemblies	3.4.3
Finish	3.13
Marking	3.14
Welding	3.20
Terminals	3.21.4

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4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspections procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit of equipment shall be inspected for conformance to all the inspection and test requirements of Table II of this specification. Equipment conforming to the above requirement will be subject to verification utilizing the procedures for multilevel continuous sampling plan (CSP-M) of standard MIL-STD-1235, unless otherwise specified.

Table II - Group A inspection

Inspection	Req't Para	Insp Para	AOQL Major	AOQL Minor
Visual and Mech.	3.23	4.16	1.5	5.0
Electrical			1%	
Power wiring			for	
Breakdown voltage	3.4.1.1	4.6.1.1	the	
Distrib.Circuits No. 1 thru No.6	3.4.1.2	4.6.1.2 through 4.6.1.7	entire	
Emergency Light	3.4.1.2	4.6.1.8	group	

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Signal wiring	3.4.2	4.6.2	of
Sys Operation AN/TTC-23()			power and signal wiring and Systems operation
Power Panel SB-1399 ()/GTA-14(V)	3.5.1	4.6.3.1	
Subsidiary functions	3.5.2	4.6.3.2 4.6.3.3 4.6.3.5 4.6.3.6 4.6.3.7 4.6.3.8 4.6.3.9	
CB-LB Line Circuits	3.5.3	4.6.3.4 4.6.3.10	
Information Turret	3.5.4	4.6.3.12	
Trunk Circuit	3.5.5	4.6.3.11	
Intercommunication Station	3.5.6	4.6.3.13	
Supervisor's Phone Operation	3.5.7	4.6.3.14	
Spare Lines on Jack Panel	3.5.8	4.6.3.15	

** All defects are major

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4.5.1.1 Group A sampling plans.- The Group A sampling plan for the AOQL's listed in Table II of this specification, when used in conjunction with Table X of Standard MIL-STD-1235, shall be as follows:

AOQL % Def	f	K	i	k 1	k 2	k 3
1.5	½	*2	43	½	*¼	Not to be used
5.0	½	*3	15	½	¼	**1/8

* Max "K" inspection level permitted
** Max "k" sampling freq permitted

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III of this specification and to the procedures for the multi-level continuous sampling plan (CSP-M) of standard MIL-STD-1235, unless otherwise specified. Group B inspection shall be performed on units of production that have passed Group A inspection. The government reserves the right to require complete retest of any sample which is reworked after failing Group B inspection.

4.5.2.1 Group B sampling plans for the AOQL's listed in Table II of this specification, when used in connection with Table X of MIL-STD-1235, shall be as follows:

AOQL% Def	f	k	i	k 1	k 2	k 3
3.0	½	*2	20	½	**¼	Not to be used
5.0	½	*3	15	½	¼	**1/8

* Max "K" insp level permitted
** Max "k" sampling freq permitted

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

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Table III.- Group B inspection

Inspection	Req't Para.	Insp Para.	AOQL % DEF (See 4.5.2.1)
Mounting insert	3.11	4.13.2	5.0
Watertightness	3.9	4.10	3.0
Fording	3.10	4.11	3.0
Interchangeability	3.22	4.15	5.0

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

Table IV.- Group C inspection

Inspection	Req't Para	Insp Para
Construction verification	3.3	4.19
Drops, flat and rotational	3.8	4.9
Solderless-terminal-lug-connection	3.21.4	4.12

4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review in accordance with paragraph 4.19, shall be performed once on each part and assembly, randomly selected, and their installation into one of first 10 shelter facilities produced.

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4.5.3.3 Sampling for inspection of drop tests.- One sample shall be selected at random from the first 50 units or fraction thereof, and one sample from each additional 150 units or fraction thereof, produced.

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.4 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective actions has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

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4.6 Electrical installation test.-

4.6.1 Power wiring.-

4.6.1.1 Insulation Breakdown Test.-

a. Remove all resistive and inductive loads from the lines; such as, 1/8A power panel fuse, fluorescent, incandescent, neon lamps, AC Voltmeter, and power plugs to equipments.

b. Turn all breakers and switches to their ON position.

c. Close the shelter door so that the blackout switch is activated.

d. Apply 500 VDC test voltage, for a period of ten (10) seconds, between terminals 1 and 4, 1 and 2, and 1 and shelter ground of the power IN receptacle. The leakage resistance shall be no less than 100 megohms.

e. After test, return all switches and breakers to their OFF positions and restore all resistive and inductive loads.

4.6.1.2 Operation of power distribution panel.- With power connected, voltmeter on panel should read nominal 115 VAC. Neon lamp shall light indicating power is connected to shelter.

4.6.1.3 Test of Circuit Breaker No.1.-

a. Place Circuit Breaker No.1, wire chief's Night Light, fluorescent, and incandescent switches in their ON position. Neon lamp shall light.

b. Place the Blackout Bypass switch in the BYPASS position. All lights shall come on.

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c. With shelter door open and the Blackout Bypass switch in the BLACKOUT position all lights shall extinguish except the two blue lights over switchboards.

d. Close shelter door and note that all lights come on except the two blue lights which extinguish.

e. Open shelter door, place Blackout Bypass switch in the BYPASS position. All lights shall come on. Leave switch in this position for remainder of tests.

f. Operate Circuit Breaker No.1 to assure lights are controlled by this breaker. Leave breaker in the ON position.

g. Turn off wire chief's Night Light and cold start incandescent switches.

4.6.1.4 Test of Circuit Breaker No.2 and No.3.-

a. Place Circuit Breaker No.2 in its ON position. Neon lamp shall light.

b. Place heater No.1 switch in the FAN position. Heater shall blow air.

c. Set thermostat to maximum temperature range.

d. Place switch in the HEAT position. After a short time the heater shall blow warm air.

e. Lower thermostat setting until a "click" is heard. This is the thermostat cutout and may be verified by observing a current drop in the power panel ammeter. The fan shall continue to operate.

f. With heater fan blowing, operate Circuit Breaker No.1 and verify it controls heater No.1. Leave breaker in the OFF position.

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g. Repeat steps a thru f for Circuit Breaker No.3 and heater No.2.

4.6.1.5 Test of Circuit Breaker No.4.-

a. Place Circuit Breaker No.4 in its ON position. Neon lamp shall light.

b. Set wall Thermostat sufficiently low to permit sensing a higher temperature.

c. Set Air Conditioner Thermostat to maximum cooling. Place switch in FAN position. Fan shall blow air.

d. Place switch to COOL position. Compressor shall come on. This can be verified by observing an increased current reading at the power panel ammeter.

e. After a short time the air conditioner shall blow cold air.

f. Increase set point of wall thermostat to a temperature higher than inside shelter temperature. Air Conditioner shall stop.

g. Reset wall thermostat to start air conditioner with air conditioner running, operate Circuit Breaker No.4 and verify it controls air conditioner. Leave breaker in the OFF position.

4.6.1.6 Test of Circuit Breaker No.5.-

a. Place Circuit Breaker No.5 in its ON position. Neon lamp shall light.

b. Remove battery exhaust plug from battery exhaust vent on the exterior of shelter.

c. Plug battery exhaust blower power cord into battery exhaust outlet. Observe that air is being exhausted through the vent to the exterior of the shelter.

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d. Operate Circuit Breaker No.5 and verify it controls the battery exhaust blower.

e. With a test lamp plugged into the Equipment Outlet on curbside wall, operate Circuit Breaker No.5 and verify it controls this outlet. Leave breaker in the OFF position.

4.6.1.7 Test of Circuit Breaker No.6.-

a. Place Circuit Breaker No.6 in its ON position. Neon lamp shall light.

b. Open Exhaust Blower louver located on bottom of blower housing on exterior of shelter.

c. Place Exhaust Blower switch in its ON position. Blower shall exhaust shelter air.

d. Operate circuit breaker and verify it controls blower.

e. Place Exhaust Blower switch in its OFF position.

f. Plug in Intercom power cord in the Intercom outlet. Turn Intercom on and note that power ON lamp lights.

g. Operate circuit breaker and verify it controls Intercom.

h. Plug a test lamp into Convenience Outlet No.1 (curbside wall) and No.2 (roadside wall). Test lamp shall light.

i. Operate Circuit Breaker No.6 and verify it controls these outlets.

j. Repeat this procedure for duplex outlet in Power Entrance box. Leave breaker in its OFF position.

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4.6.1.8 Emergency Light Test.-

a. Install 48V batteries or their equivalent.

b. Place Emergency Light circuit breaker located in conduit between Power Distribution Panel and SB-1399/GTA-14 in its ON position.

c. Throw main breaker to OFF (Simulates loss of 110 VAC power). All lights shall extinguish and the emergencylights shall come on.

d. Place Emergency Light Switch Circuit Breaker in its OFF position. Throw main breaker ON. Emergency lights shall remain extinguished. Leave main breaker in its ON position.

4.6.2 Signal wiring.-

4.6.2.1 Insulation breakdown test.-

a. Remove all plugs from the Spares and Intercom jacks at the supervisor's desk.

b. Apply 500 VDC test voltage, for a period of ten (10) seconds, between the following points in the Signal Entrance Box:

(1) Each terminal of Signal Connector No.1 and shelter ground.

(2) Each terminal of Signal Connector No.1 and each remaining terminal within the connector.

(3) Repeat steps (1) and (2) for the five remaining signal connectors.

(4) Each PHONE BINDING POST and shelter ground.

(5) Each PHONE BINDING POST pair.

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(6) Each INTERCOM BINDING POST and shelter ground

(7) Each INTERCOM BINDING POST pair.

(8) Each INFO BINDING POST and shelter ground.

(9) Each INFO BINDING POST pair.

4.6.3 Systems Operation Test.-

4.6.3.1 Power Panel Test, SB-1399()/GTA-14(V).-

a. Plug in SB-1399()/GTA-14(V) to equipment receptacle, and place circuit breaker in its ON position.

b. Turn on 48V breaker on top panel. There shall be no indication of alarms, buzzers, etc., within the shelter.

c. Turn on 110 VAC breaker on top panel.

d. Turn FINE AND COURSE knobs on lower panel to their lowest settings.

e. Turn on AC INPUT breaker on lower panel. Pilot light shall come on.

f. Turn on the DC OUTPUT breaker on lower panel.

g. Adjust the COURSE and FINE knobs until 48 volts is obtained on the meter.

h. Turn on telering breakers on top panel and Emergency-Ring-Telering Breaker to Telering.

i. Measure for approximately 48 VDC between any fuse and ground at each trunk and live relay rack. Note that ground is positive.

j. Measure for approximately 48 VDC between any fuse and ground at the switchboards.

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(k) Measure for approximately 100 VAC, at the 20 cycle ringing plugs connected to each switchboard.

(l) Measure for approximately 115 VAC, at the AC plugs connected to each switchboard.

(m) Check each heater in switchboards to verify they are heating.

(n) Carefully depress the tabs of each fuse on the Line Relay Rack, Trunk Relay Rack, and switchboards. Buzzer alarm shall sound when each tab is depressed.

4.6.3.2 Supervisory Lamps Test.-

a. Place ANS cord plug half-way into "recall" jack. Supervisory lamp shall light.

NOTE: The "(Recall Jack)" is located in the lower right corner of switchboard face.

b. Insert ANS cord plug completely into "recall" jack. Supervisory lamp shall extinguish.

c. Repeat these two steps for all ANS and CALL cords. All lamps shall function.

4.6.3.3 Dial test.-

a. Place first RING-ANSWER-DIAL switch, starting from left of No.1 switchboard, in its locked position. (No. 1 switchboard is unit nearest door end; No.2 unit is nearest front wall.)

b. With an ohmmeter set for R x 1 scale reading, place meter leads between sleeve and ground of first (from left) CALL cord plug.

c. Dial Zero and hold the dial at this position. Ohmmeter shall not exceed 8 ohms.

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d. Release dial. When dial reaches zero, ohmmeter shall read infinity.

e. Return switch to its normal position.

f. Repeat steps (a) through (e) for all 15 plugs.

g. Repeat steps (a) through (f) for Switchboard No.2.

h. With ohmmeter set for R x 10 scale reading, place meter leads between tip and ring of first (from left) CALL cord plug of No.1 Switchboard.

i. Place RING-ANS-DIAL switch in its locked position. A reading of 500 ohms (approximately 10%) shall be obtained.

j. With leads still across tip and ring, dial zero and hold dial at this position. A reading of not more than 10 ohms is acceptable.

k. Repeat steps h through j for all 15 plugs.

l. Repeat steps h through k for Switchboard No.2.

m. With ohmmeter set for R x 10 scale, place meter leads between tip and ring of any CALL cord plug on No.1 Switchboard. Operate ring answer dial switch to locked position. Throw DIAL-WIPEOUT switch to WIPEOUT position. A reading of infinity shall be obtained.

n. With ohmmeter set for R x 1 scale, place meter leads across sleeve and ground of any CALL cord plug and a ground. A reading of not more than 8 ohms shall be obtained when DIAL-WIPEOUT switch is thrown to the WIPEOUT position.

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o. Repeat steps m and n for Switchboard No.2.

4.6.3.4 CB Line Circuit Test.-

a. Place all CB-LB switches on the line relay rack in their CB position.

b. Place NA switch in the ON position at each line relay rack.

c. Connect a CX-4566()/G cable between one Distribution Box J-1077()/U and Signal Connector No.1 in the Signal Entrance Box. Make this same cable hookup using the remaining Distribution Box J-1077()/U and Signal Connector No.3.

d. Connect a TA-312()/PT phone arranged for CB operation to binding post pair No.1 of one J-1077()/U and another TA-312()/PT phone arranged for CB operation to pair No.1 of the other J-1077()/U. These phones will be referred to as Phone No.1 and No.2 respectively.

e. Connect a headset to one of the headset receptacles on the face of the key, shelf at operator's position No.1, (nearest entrance door). Connect a headset to operator's position No.2, (nearest front wall) in a similar manner.

f. Lift Phone No.1. Line Lamp No.1 on face equipment shall light. The night alarm shall ring. Note: The Night Alarm may be muted since repeated operation will occur during test of all lines.

g. Insert ANS cord into jack associated with the lighted line lamp. The lamp shall extinguish and the Night Alarm shall stop ringing.

h. Place RING-CALL-TALK switch in the ANS position. Communication through Phone No.1 and Operator's headset shall be loud and clear.

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i. Repeat sequence of steps f,g and h above for Phone 2 except use CALL cord. Communication shall be loud and clear.

j. Replace Phone No.1 and No.2 in their cradles. The CALL SUPERVISORY LAMP and ANS SUPERVISORY LAMP shall light.

k. Momentarily operate both RING CALL-TALK and RING ANS-DIAL switches to their RING positions. Phone No.1 and No.2 shall ring, respectively. Call cord must be in.

l. Withdraw both plugs from the jacks in the face equipment. All lamps shall extinguish.

m. Repeat steps f through l at Operator's position No.2.

n. Repeat steps f through m for all lines (20). Use all ANS and CALL cords (15) at each Operator's position until each is verified as satisfactory.

o. After completion of test for Signal Connector No.1 and No.3, move cables to Signal Connector No.2 and No.4 and repeat steps f through n with the following exceptions:

(1) Use of all ANS and CALL cords is not required.

(2) At any convenient time, substitute the remaining four (4) operator's headsets for verification of their satisfactory operation.

(3) At any convenient time, utilize the remaining two (2) headset receptacles on the face of the key shelf.

(4) At any convenient time, temporarily place TELERING EMERGENCY RING breaker to the EMERGENCY RING position. Turn on energy ring circuit breaker. A complete operation sequence of the two phones shall be possible.

p. After completion of test for Signal Connectors No.2 and No.4, connect a cable between Signal Connector No.5 and a J-1077()/U Distribution Box.

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q. Connect Phone No.1 to binding post pair No.1 and Phone No.2 to binding post pair No.11 of the J-1077()/U.

r. Repeat steps f through n above for Signal Connector No.5.

s. Leave all connections unchanged after completing test.

4.6.3.5 Hand Ringing Generator Test.-

a. Place the RINGING-KEY-HAND switch on Switchboard No.1, face equipment to the HAND position.

b. Insert CALL cord plug into jack No.1 (Signal Connector No.5). Lift handset of phone No.1 momentarily to assist operator in locating of jack.

c. Place the RING-CALL-TALK switch in the RING position.

d. Turn hand crank on the switchboard shelf. Phone No.1 shall ring. Return RINGING-KEY-HAND switch to KEY position and remove plug.

e. Repeat steps a through d for Switchboard No.2. Leave all connections unchanged after completing test.

4.6.3.6 Monitor-Master Ringing Test.- For party line

a. Operate the first from left RING-CALL-TALK switch to the RING position.

b. Place MONITOR-MASTER RINGING switch in its locked position.

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c. Measure voltage between the tip of the associated call cord and a ground. A reading of 100 VAC shall be obtained.

d. Repeat the above using all RING-CALL-TALK switches and associated CALL cords on switchboards 1 and 2.

4.6.3.7 Position Grouping Test.-

a. Lift handset of Phone No.2. Lamp at jack No. 11 on face equipment shall light.

b. Insert ANS cord plug into this jack at Switchboard No.2. Lamp shall extinguish.

c. Place RING-CALL-TALK switch to TALK position.

d. Place POSITION-GROUPING switch in its locked position.

e. Communication between Phone No.2 and Switchboard No.2 operator shall be loud and clear. Remove cord plug and return switches to normal position. Leave all connections unchanged after completing test.

4.6.3.8 Monitor test.-

a. Lift handset of Phone No.1. Lamp at jack No.1 on face equipment shall light.

b. Insert ANS cord plug into this jack at Switchboard No.1.

c. Place RING CALL-TALK switch to the TALK position.

d. Operator at switchboard shall give a long count. Phone No.1 operator shall not be able to hear long count when MONITOR-MASTER-RINGING switch is placed in the MONITOR position by the switchboard operator.

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e. Remove ANS cord plug and return switches to normal position.

f. Repeat steps a through f for Switchboard No.2.

4.6.3.9 Conference Call Test.-

a. Lift handset of Phone No.1 lamp at jack on face equipment shall light.

b. Insert ANS cord plug into this jack at switchboard No.1.

c. Place RING-CALL-TALK switch to the TALK position.

d. Phone No.1 operator and switchboard operator shall be in communication.

e. Place CALL cord plug into first conference jack (lowest row, right half of face equipment) at switchboard No.1.

f. Insert second ANS cord plug into second conference jack.

g. Lift handset of Phone No.2 and insert CALL cord plug associated with second ANS cord into jack on face equipment that lights. Place second RING-CALL-TALK switch in TALK position; communication between switch board and phones and between phones shall be loud and clear.

h. Place RING-CALL-TALK switches in normal position. Communication between phones still exists.

i. Remove plug from second conference jack and place in third conference. Communications between phones still exists. Place RING-CALL-TALK switch in TALK position; communication between switchboard and phones still exists.

j. Repeat repositioning of latter cord plug until all 19 conference jacks are verified for load and clear communication.

k. Repeat steps a through j for switchboard No.2.

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4.6.3.10 LB Line Circuit Tests.-

a. Place all CB-LB switches on the line relay rack on their LB positions.

b. Arrange Phone No.1 and Phone No.2 for LB operation.

c. Perform the same sequence of operations as outlined in Paragraph 4.6.3.4 with the following differences:

(1) Both phones will require hand ringing to signal switchboard and for ringdown when calls are completed.

(2) Magneto supervisory lamps will light in lieu of supervisory ANS and CALL lamps when ringdown is accomplished at Phone No.1 and No.2.

4.6.3.11 Trunk Circuit Tests.-

a. Throw all idle lamp control switches on TA-276()/TTC rack to ON position. Each idle trunk lamp on face equipment shall light.

b. Reposition switches so only lights one (1) and eleven (11) on both switchboards remain on.

c. Arrange idle trunk hookup as shown in Figure #1.

d. Place first (extreme left) CALL cord plug of Switchboard No.1 into trunk line jack No.11. Idle lamp for jack No.11 shall go out and adjacent lamp for jack No.12 shall light. (Both switchboards).

e. Place ANS cord plug of Switchboard position No.2 into trunk line jack No.1. Idle lamp shall extinguish and second idle lamp shall light. (Both switchboards).

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f. Place RING-CALL-TALK keys of both switchboards in their TALK positions. Communication between switchboard headsets shall be loud and clear. Return keys to neutral positions.

g. Temporarily withdraw CALL cord plug from jack. Idle lamp (No.12) shall go out and lamp at position No.11 shall come on. ANS supervisory lamp at Switchboard position No.2 shall light.

h. Withdraw ANS cord plug at Switchboard position No.2. The ANS supervisory lamp shall go out.

i. Reinsert CALL cord plug into jack No.11 to simulate an "in use" trunk.

j. Repeat steps a through i utilizing CALL cord plugs sequentially, 12 through 20 on Switchboard No.1 and move connections at J-1077()/U to pairs 2 and 12, 3 and 13 etc.

k. Repeat sequence outlined in steps d through j for Switchboard position No.2.

l. Return hookup to conform to Figure 1 and remove all plugs.

m. Repeat d through j using CALL cords at Switchboard No.1, trunks 1 through 10 and ANSWER cords at Switchboard No.2, trunks 11 through 20.

n. Repeat d through j using CALL cords at Switchboard No.2, trunks 1 through 10 and ANSWER cords at Switchboard No.1, trunks 11 through 20.

4.6.3.12 Information Turret (Info phones) Test.-

a. Place cross connection between pairs 98 and 99 of trunk Main Distribution Frame. To line main distribution frame.

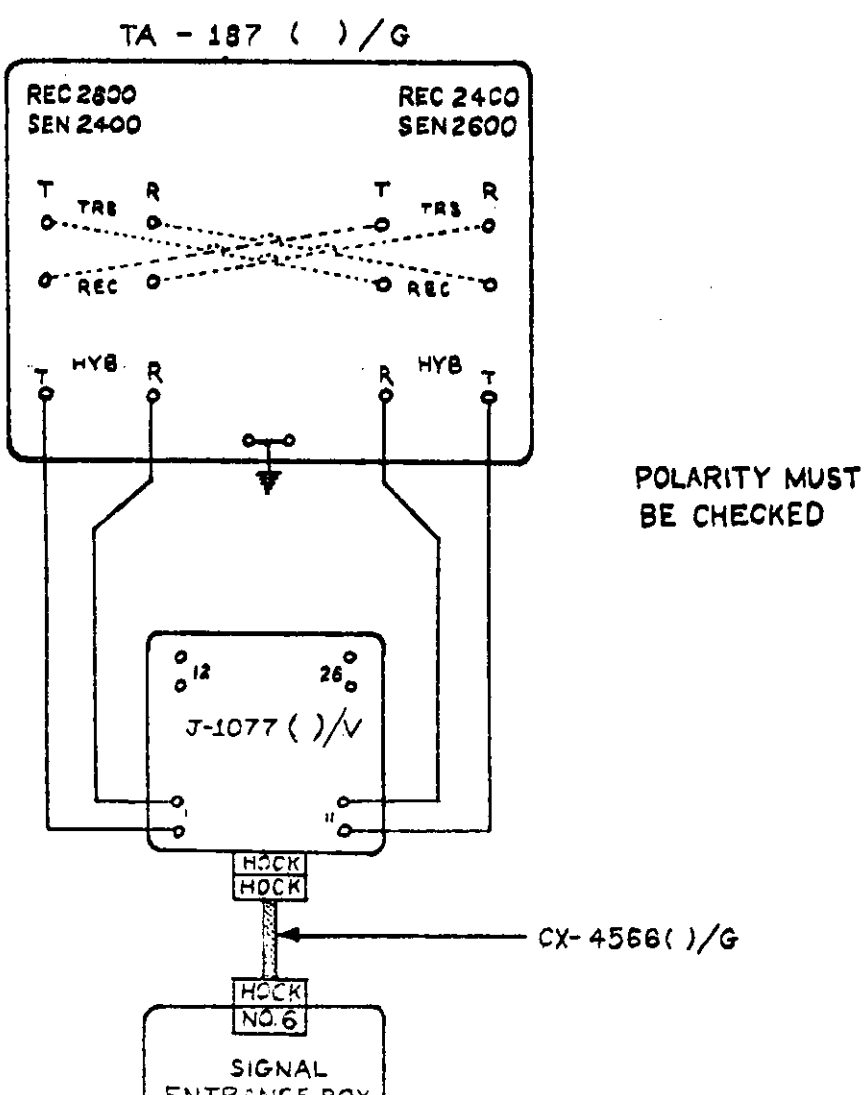
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FIGURE 1

IDLE LAMP CONTROL TRUNK TEST PER SPECIFICATION MIL - C - 55537A(EL)

TRUNK HOOKUP

TA - 187 ( )/G



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b. Originate a call from Phone No.1 on a pair at the J-1077()/U. (Still connected to No.5 signal connector).

c. Switchboard operator accepts call with ANS cord plug.

d. Operator inserts CALL cord plug into Information Line No.1 jack on face equipment. (Corresponds to pair 98).

e. Operator rings Information Turret Position No.1 at supervisor's desk. CALL light on information turret shall blink when operator rings position.

f. Place Information Turret Switch in TALK position. Establish communication between phone outside shelter and Info Phone 1 at desk.

g. Place Info Turret Switch Line No.1 in HOLD position. Phone No.1 and Supervisor's position shall not be capable of communicating. HOLD lamp shall light.

h. Originate a call from Phone No.2 at the J-1077()/U.

i. Operator places Phone No.2 through to Information Line No.2. Phone No.1 shall not hear conversation.

j. Place Information Line No.2 in HOLD position. HOLD lamp shall light. Place info Turret switch in TALK position. Operator and Phone No.1 shall resume conversation. Phone No.2 shall not hear conversation.

k. Replace Info Turret switches to NORMAL and withdraw switchboard cords.

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4.6.3.13 Intercommunication station test.-

a. With a suitable low voltage indicating device, such as a buzzer, check for continuity between terminals 26A and 26B of each connector (1 through 6) and the binding posts marked INTERCOM in the Signal Entrance Box.

b. Connect an LS-147()/FI Intercommunication Station to the INTERCOM Binding Posts.

c. Plug the LS-147()/FI at the Supervisor's desk into the INTERCOM jack.

d. Apply power to both INTERCOM units. Power ON lamps shall light.

e. Communications between INTERCOM units shall be possible.

4.6.3.14 Supervisor's phone test.-

a. With a suitable low voltage indicating device, such as a buzzer, check for continuity between terminals 25A and 25B of each connector (1 through 6) and the Binding Posts marked PHONE (Supervisor) in the Signal Entrance Box.

b. Connect a TA-312()/PT phone arranged for LB operation to the Binding Posts marked PHONE in the Signal Entrance Box.

c. Arrange the supervisor's phone for LB operation.

d. Plug the supervisor's phone into the jack marked PHONE; loud and clear communication shall be possible between the two phones.

4.6.3.15 Spare phone jacks test.-

a. Using a low voltage indicating device, verify that pairs 21 through 24 are in parallel in all six signal connectors.

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b. Connect the J-1077()/U to one of the signal connectors in the entrance box.

c. Connect a telephone set for LB operation to pair 21 of the J-1077()/U.

d. Connect the supervisor's phone set for LB operation to spare jack 21 and establish communications with outside phone.

e. Repeat for remaining spare lines using jacks and corresponding pairs in the J-1077()/U.

4.7 Vehicular transportation test.- The Central Office, Telephone, Manual AN/TTC-23() shall be secured to a Government-furnished Truck, 2½ ton, M-35 and transported over the specially prepared course described in 4.7.1 for 10 laps at 5 mph, 10 laps at 10 mph, 10 laps at 15 mph, and 10 laps at 18 to 20 mph. A lap is defined as traversing the course in one direction. The AN/TTC-23() shall meet the requirements of paragraph 3.6 after completion of the test.

4.7.1 Vehicular transportation course. The course for the military truck transportation test shall have twelve (12 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1 inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be anchored securely by any convenient means.
The ground on which the timbers are placed shall be hard and level.

NOTE: Battery Dummy-Loads shall be installed for this test.(See 6.2(b).

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4.8 Railroad transportation test.- The Central Office, Telephone, Manual AN/TTC-23() shall be loaded on a railroad flatcar (test car) in accordance with Figure 2. The test shall be conducted on a flat stretch of track. A loaded car totaling 165,000 lbs., traveling at 9 miles per hour (or equivalent impact energy conditions approved by the Government) shall be impacted against the test flat car coupled to two other empty cars. These three cars shall be stationary with brakes off. Four impacts shall be performed, two impacts into each end of the test flatcar. The cabling and blocking holding the shelter in place shall be tight at the start of each impact run. If the cabling or blocking is torn loose by the impact, the run shall be repeated. A damage survey shall be made after each impact and the AN/TTC-23() shall meet the requirements of 3.7.

NOTE: The tests shall be conducted without Batteries (or equivalent) installed.

4.9 Drop tests.- These tests shall be performed using Battery Dummy-Loads. (See 6.2(b)).

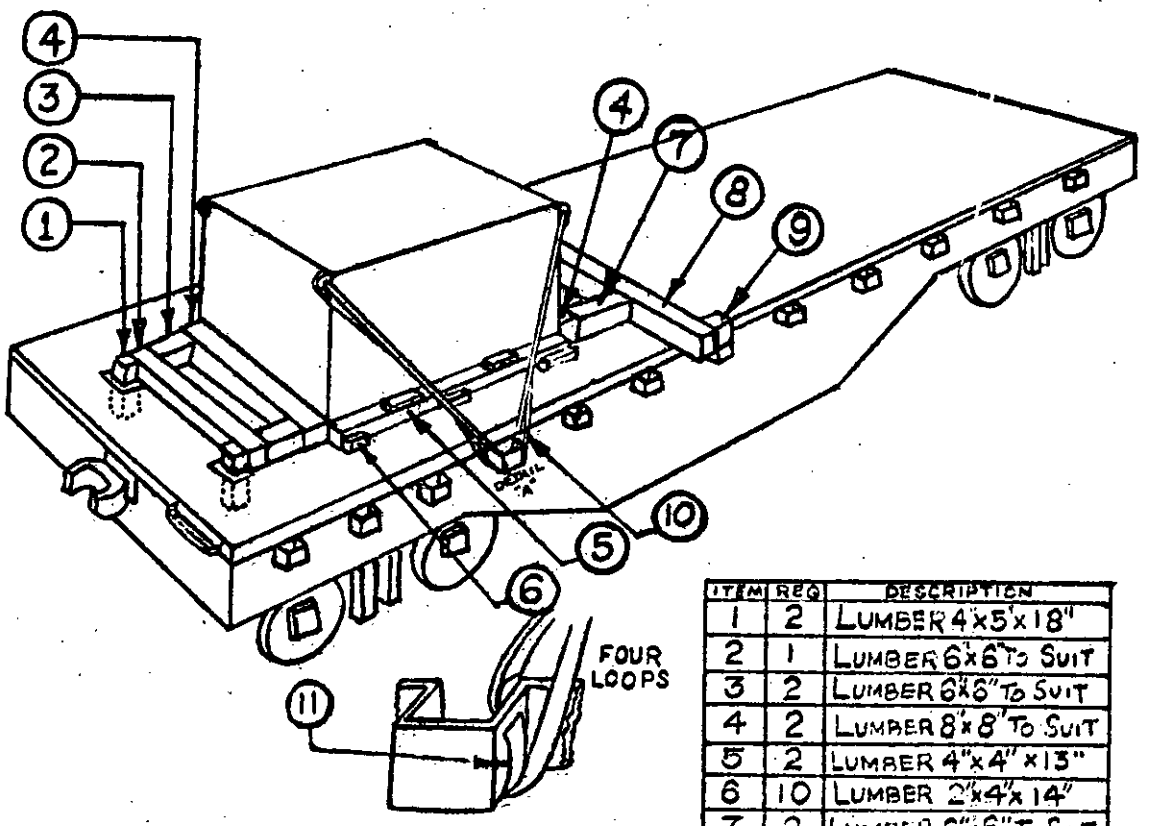
4.9.1 Flat drop.- The Central Office, Telephone, Manual AN/TTC-23() shall be lifted and allowed to fall freely a distance of 18 inches with the skids impacting onto a hard concrete surface. This test shall be performed once. Upon completion of the test, the AN/TTC-23() shall meet the requirements of 3.8.

4.9.2 Rotational drops.- The Central Office, Telephone, Manual AN/TTC-23() shall have one edge placed on a nominal 4-inch high timber. The opposite edge of the shelter shall be raised 18 inches and allowed to fall freely onto a hard concrete surface. This drop test shall be performed on each bottom edge of the shelter. Upon completion of these drops, the shelter shall meet the requirements of 3.8. In each drop test, the 4-inch high member shall be placed under the skids and the eighteen inch measurement shall be taken from the bottom of the skids.

4.10 Watertightness.- The shelter, without the use of any additional caulking, sealing, taping, etc., shall be subjected to the following tests:

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FIG 2



WIRE TIEDOWN SAME FOR ALL POINTS OF CONTACT WITH STAKE POCKETS

ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8" TO SUIT
5	2	LUMBER 4"x4" x13"
6	10	LUMBER 2"x4"x 14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	16	WIRE #9 SOLID, HARD 2000 LB TEST
11	4	FLEX METAL STRAPPING

NOTES:

A. THE BLOCKING LUMBER SHALL BE NOTCHED TO ACCOMODATE TOWING EYES, BOLTS AND RIVET HEADS
B. THE BLOCKING LUMBER SHALL BE RIGIDLY FASTENED TO THE FLOOR OF THE FLATCAR

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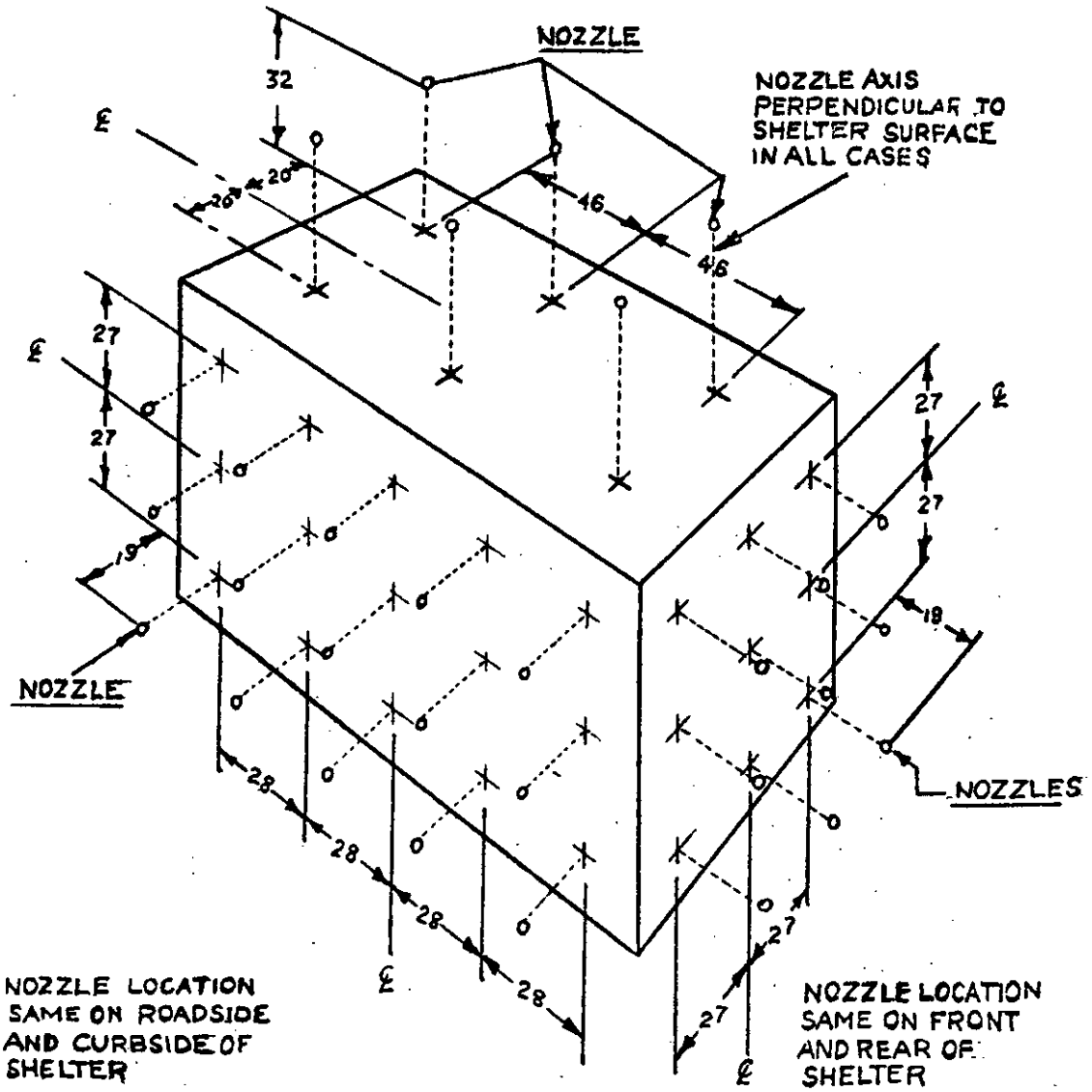
a. Each of the five exposed surfaces of the shelter, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches and the row of nozzles in line with the air conditioner where the 19 inches shall be measured from the extreme vertical surface of the air conditioner. (See Figure 3. for nozzle location.) Each panel shall be sprayed for a period of twenty minutes. (If the watertightness test is to be followed by the fording test of 4.11, the bottom row of nozzles, with the exception of the nozzles facing the door-end panel, can be eliminated.) Upon completion of the above test, the AN/TTC-23() shall meet the requirements of 3.9. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position and repeat a. above.

4.11 Fording test.- The shelter, prepared for transportation, and without any additional sealer, caulking, taping, fording kit, etc., shall be immersed in water to a depth of 21 inches, measured from the bottom of the skids, for a period of one hour. The shelter shall be restrained by external tiedowns. Upon completion of this test, the shelter shall meet the requirements of 3.10. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

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FIGURE 3
NOZZLE LOCATION FOR TEST PER SPECIFICATION
MIL-C-55537A(EL)



NOTE:
GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES

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4.12 Testing of solderless terminal-lug connections.- Connections made with solderless terminal lugs shall be tested to meet the requirements of 3.21.4.

4.13 Mounting insert tests.-

4.13.1 Preproduction insert test.- Four of each size insert shall be randomly selected for test. Each insert selected shall be tested with an axial load followed by a torque load in accordance with Table V. After test the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection.

4.13.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than 5, all of that size shall be tested) shall be randomly selected in each sample equipment. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample equipment.

Table V. Mounting Insert Loads

Insert size	Preproduction Test		Production Test	
	Axial(lbs)	Torque(in-lbs)	Axial(lbs)	Torque(lbs)
1/4-20	1300	60	650	60
5/16-18	2000	100	1000	100

4.14 Sealer shear tests.- A tensile pull test shall be performed on sample lap joints consisting of two (2) one-inch wide strips of aluminum (0.062 inch thick) overlapped one-half inch, giving contact area of one-half square inch and bonded with sealer material over no more than the specified one-half square inch area. The samples shall be tested over the conditions of 4.14.1 through 4.14.3. Five specimens shall be used for each test condition and tests shall be to destruction with rate of application of load constant at one inch per minute maximum.

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4.14.1 Shear at temperature.- Five samples shall be subjected to a tensile pull test at ambient temperature (80°F ± 15°); five samples at -65°F ± 5°; and five samples at 200°F ± 5°. Samples shall meet respective requirements of 3.16.1. Temperature of sample shall be stabilized at specified level as indicated by a thermocouple at the bond area.

4.14.2 Shear after humidity exposure.- Each sample shall be tested after exposure to 100% Relative Humidity at 160°F ± 5° for two weeks and shall meet requirements of 3.16.2. Temperature of sample shall be stabilized at specified level when tested.

4.14.3 Shear after salt spray exposure.- Each sample shall be tested for conformance to 3.16.3 after being subjected to the salt spray test of Standard MIL-STD-202 using a 20 percent NaCl solution at 95°F for a period of 2 weeks.

4.15 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.22. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets.

c. Size and mounting dimensions of equipment drawers.

d. Size, mounting dimensions, and capacity of ventilating equipment.

4.16 Visual and Mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.23 and Table VI.

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Table VI. Classification of Visual & Mechanical Defects

Classification	Defects
Major	1. Mounting insert missing or mislocated. 2. Sealer improperly applied (holes,separations, or lack of adhesion). 3. Welds - cracked or porous. 4. Parts, brackets, or equipment loose or missing. 5. Drawers, doors, etc., inoperative 6. Blower inoperative or rotating in wrong direction. 7. Mounting plates missing or mislocated. 8. Aircraft loading data plate, instructions or nameplate omitted, incorrect or illegible. 9. Power receptacles, switches, jacks, wire ducts or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Lock washer or lock nut missing. 12. Burrs and sharp edges in wire duct not removed. 13. Incorrect or missing labels on Binding Posts and jacks.
Minor	1. Screw loose. 2. Doors, slides, drawers not easily operated. 3. Finish-abrasions or scratches. 4. Locking or holding devices difficult to operate. 5. Burrs and sharp edges on brackets not removed.

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4.17 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspection in accordance with Specification MIL-P-55505.

4.18 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph 4.6.3	Systems operation
Paragraph 4.16	Visual and mechanical

4.19 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawings cited in the Technical Data Package (as supplemented by approved Engineering Change Proposals (ECP) resulting from Production Evaluation). EVR shall consist of the following:

a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in sub- assemblies and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, component, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information

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(1) Identification of parts, components, shelter facilities and details proving compliance with 4.a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Two (2) copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the Government technical organization cited elsewhere in the contract for submission of ECP's. The results of the EVR shall be submect to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to Government personnel during the inspection. The contractor shall provide the following:

(1) Segregation of parts, components of shelter facilities to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have, as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the AN/TTC-23() shall be in accordance with applicable requirement of Specification MIL-S-55505.

6. NOTES

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6.1 Intended use.- The Central Office, Telephone, Manual AN/TTC-23( ) is intended for use by division and will provide one multipled two-position switching facilities for a division area type communications system.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Preproduction and Group C Inspection of the AN/TTC-23( ) requires contractor fabricated dummy loads for the BB-46 Batteries (used in the vehicular transportation and drop tests) unless otherwise specified in the battery hold-down bracketry.

c. A list of each individual component item to be used for testing the equipment.

d. Submission of the statement of treatment referenced in 3.19.2 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (See 4.5.3.3)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: AN/TTC-23( ) becomes AN/TTC-23A. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army (EL)

Preparing Activity
Army - EL
Project No. 5805-A099

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-55537A(EL) CENTRAL OFFICE TELEPHONE, MANUAL AN/TTC-23( )

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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